Section 4.1

2. a. We can know the uniform distribution $f(x;-5,5)=\begin{cases}\frac{1}{10}, & -5\le x\le 5\\ 0, & \text{otherwise.}\end{cases}$

$P(X<0)=\int_{-5}^{0} f(x)\,dx=\left[\frac{1}{10}x\right]_{-5}^{0}=\frac{1}{2}=0.5$

b. $P(-2.5<X<2.5)=\int_{-2.5}^{2.5} f(x)\,dx=\left[\frac{1}{10}x\right]_{-2.5}^{2.5}=0.5$

c. $P(-2\le X\le 3)=\int_{-2}^{3} f(x)\,dx=\left[\frac{1}{10}x\right]_{-2}^{3}=0.5$

d. $P(k<X<k+4)=\int_{k}^{k+4} f(x)\,dx=\left[\frac{1}{10}x\right]_{k}^{k+4}=\frac{1}{10}(k+4-k)=0.4$

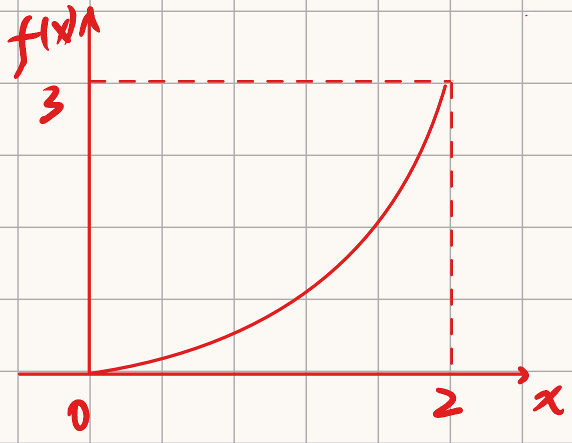
5. a. By the properties of pdf, we have

$$\int_{-\infty}^{\infty} f(x)\,dx=\int_{0}^{2} kx^2\,dx=\left[\frac{1}{3}kx^3\right]_{0}^{2}=\frac{8}{3}k=1.$$

Thus $k=\frac{3}{8}$.

$$f(x)=\begin{cases}\frac{3}{8}x^2, & 0\le x\le 2.\\ 0, & \text{otherwise.}\end{cases}$$

Density curve:



b. $P(X\le 1)=\int_{-\infty}^{1} f(x)\,dx=\int_{0}^{1}\frac{3}{8}x^2\,dx=\left[\frac{1}{8}x^3\right]_{0}^{1}=\frac{1}{8}=0.125$

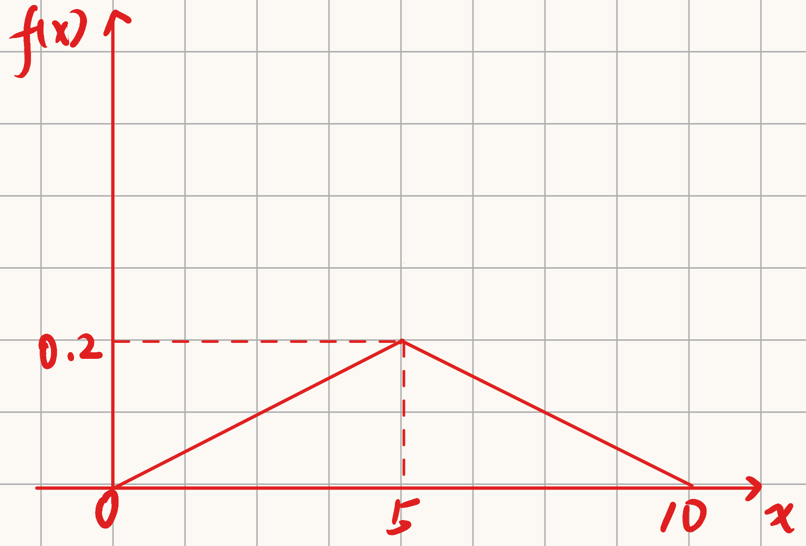
c. 60 sec = 1 min, 90 sec = 1.5 min

Thus we calculate

$$P(1 \le X \le 1.5) = \int_1^{1.5} f(x)\,dx = \left[\frac{1}{8}x^3\right]_1^{1.5} = \frac{19}{64} = 0.297$$

d. $P(X \geq 1.5) = \int_{1.5}^{\infty} f(x)\,dx = \int_{1.5}^{2} \frac{3}{8}x^2\,dx = \left[\frac{1}{8}x^3\right]_{1.5}^{2} = \frac{37}{64} = 0.578$

8. a.



b. $\int_{-\infty}^{\infty} f(y)\,dy = \int_0^5 \frac{1}{25}y\cdot dy + \int_5^{10} \left(\frac{2}{5} - \frac{1}{25}y\right) dy$

$= \left[\frac{1}{50}y^2\right]_0^5 + \left[\frac{2}{5}y - \frac{1}{50}y^2\right]_5^{10}$

$= 0.5 + 0.5$

$= 1$

c. Let X = the total waiting time.

$$P(X \le 3) = \int_{-\infty}^{3} f(y)\,dy = \int_0^3 \frac{1}{25}y\,dy = \left[\frac{1}{50}y^2\right]_0^3 = 0.18$$

d. $P(X \le 8) = \int_{-\infty}^{8} f(y)dy = \int_0^5 \frac{1}{25}y\,dy + \int_5^8 (\frac{2}{5} - \frac{1}{25}y)dy$

$= [\frac{1}{50}y^2]_0^5 + [\frac{2}{5}y - \frac{1}{50}y^2]_5^8$

$= 0.92$

e. $P(3 \le X \le 8) = \int_3^8 f(y)dy = \int_3^5 \frac{1}{25}y \cdot dy + \int_5^8 (\frac{2}{5} - \frac{1}{25}y)dy$

$= [\frac{1}{50}y^2]_3^5 + [\frac{2}{5}y - \frac{1}{50}y^2]_5^8$

$= 0.74$

f. Let p denote as the probability.

$p = P(X \le 2) + P(X \ge 6) = \int_0^2 \frac{1}{25}y \cdot dy + \int_6^{10} (\frac{2}{5} - \frac{1}{25}y)dy$

$= [\frac{1}{50}y^2]_0^2 + [\frac{2}{5}y - \frac{1}{50}y^2]_6^{10}$

$= 0.4$

Section 4.2

12. a. $P(X<0) = F(0) = \frac{1}{2} + \frac{3}{32}(4 \cdot 0 - \frac{0^3}{3}) = 0.5$

b. $P(-1<X<1) = F(1) - F(-1) = \frac{3}{32}[4 \times 1 - \frac{1}{3} - 4 \times (-1) + \frac{(-1)^3}{3}]$

$= \frac{11}{16} = 0.6875$

c. $P(0.5<X) = 1 - P(X \leq 0.5) = 1 - F(0.5) = 1 - \frac{175}{256} = 0.3164$

d. When $x<-2$ or $x \geq 2$, $f(x) = F'(x) = 0$.

When $-2 \leq x < 2$, $f(x) = F'(x) = \frac{3}{8} - \frac{3}{32}x^2$

Thus $f(x) = \begin{cases} \frac{3}{8} - \frac{2}{32}x^2, & -2 \leq x < 2 \\ 0, & \text{otherwise.} \end{cases}$

e. $F(\tilde{\mu}) = \frac{1}{2} + \frac{3}{32}(4\tilde{\mu} - \frac{\tilde{\mu}^3}{3}) = 0.5$.

Thus $4\tilde{\mu} - \frac{\tilde{\mu}^3}{3} = 0$. Thus $\tilde{\mu} = 0$ or $\pm 2\sqrt{3}$ (illegal)

Thus $\tilde{\mu} = 0$.

17. a. $f(x) = \begin{cases} \frac{1}{B-A}, & A \leq x \leq B \\ 0, & \text{otherwise.} \end{cases}$

$F(x) = \int_{-\infty}^{x} f(y)\,dy = \int_{A}^{x} \frac{1}{B-A}\,dy = [\frac{y}{B-A}]_{A}^{x} = \frac{x-A}{B-A}$

Thus $F(x) = \begin{cases} 0, & x<A \\ \frac{x-A}{B-A}, & A \le x \le B \\ 1, & x>B \end{cases}$

The expression

$$p = F(\eta) = \begin{cases} 0, & \eta<A \\ \frac{\eta-A}{B-A}, & A \le \eta \le B \\ 1, & \eta>B. \end{cases}$$

Thus $\eta(p) = (B-A)p + A$, $A \le \eta \le B$.

b. $E(x) = \int_{-\infty}^{\infty} x f(x) dx = \int_A^B \frac{x}{B-A} dx = \left[\frac{1}{2(B-A)} x^2\right]_A^B = \frac{B+A}{2}$.

$$V(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx = \int_A^B (x - \frac{B+A}{2})^2 \frac{1}{B-A} \cdot dx$$

$$= \frac{1}{B-A}\left[\frac{1}{3}x^3 - \frac{B+A}{2}x^2 + \left(\frac{B+A}{2}\right)^2 \cdot x\right]_A^B$$

$$= \frac{1}{3}(B^2+AB+A^2) - \frac{1}{4}(B^2+2AB+A^2)$$

$$= \frac{1}{12}A^2 + \frac{1}{12}B^2 - \frac{1}{6}AB.$$

$$= \frac{1}{12}(A-B)^2$$

$$\sigma_x = \sqrt{V(X)} = \sqrt{\frac{1}{12}(A-B)^2} = \frac{\sqrt{3}}{6}(B-A).$$

c. $E(X^n) = \int_{-\infty}^{\infty} x^n f(x) dx = \int_A^B \frac{x^n}{B-A} dx = \left[\frac{x^{n+1}}{(n+1)(B-A)}\right]_A^B$

$$= \frac{B^{n+1} - A^{n+1}}{(n+1)(B-A)}$$

22. a. $F(x) = \int_{-\infty}^{x} f(y)\,dy = \int_{1}^{x} (2 - \frac{2}{y^2})\,dy = [2y + \frac{2}{y}]_1^x$

$= 2x + \frac{2}{x} - 4$

Thus $F(x) = \begin{cases} 0, & x < 1 \\ 2x + \frac{2}{x} - 4, & 1 \le x \le 2. \\ 1, & x > 2. \end{cases}$

b. $p = F(\eta) = 2\eta + \frac{2}{\eta} - 4, \quad 1 \le \eta \le 2.$

$\begin{cases} 0, & \eta < 1 \\ 1, & \eta > 2. \end{cases}$

Thus $\eta(p) = \frac{4 + p \pm \sqrt{p^2 + 8p}}{4}$. since $1 \le \eta \le 2$.

$\eta(p) = \frac{4 + p + \sqrt{p^2 + 8p}}{4}$

when $p = 0.5$, $\tilde{\mu} = \eta(0.5) = 1.640$

c. $E(X) = \int_{-\infty}^{\infty} x f(x)\,dx = \int_{1}^{2} (2x - \frac{2}{x})\,dx = [x^2 - 2\ln x]_1^2$

$= 1.614$

$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx = \int_{1}^{2} (2x^2 - 2)\,dx = [\frac{2}{3}x^3 - 2x]_1^2$

$= \frac{8}{3}.$

$V(X) = E(X^2) - [E(X)]^2 = 0.063.$

d. Let $h(x)$ be the amount left when demand is x.

$$h(x) = \begin{cases} 1.5 - x, & 1 \le x \le 1.5 \\ 0, & 1.5 < x \le 2 \\ 2, & \text{otherwise} \end{cases}$$

$$E(h(x)) = \int_{-\infty}^{\infty} f(x) \cdot h(x) \cdot dx$$

$$= \int_1^{1.5} 2(1.5-x)(1-\frac{1}{x^2})dx + \int_{1.5}^{2} 0 \cdot f(x) \cdot dx + \int_{-\infty}^{1} \cdot 0 \cdot h(x) dx$$

$$+ \int_2^{\infty} 0 \cdot h(x) dx$$

$$= \int_1^{1.5} 2(1.5-x)(1-\frac{1}{x^2})dx$$

$$= 2\left[1.5x + \frac{1.5}{x} - \frac{1}{2}x^2 + \ln x\right]_1^{1.5}$$

$$= 0.061$$

23. Let x denote temperature in °C.

Then $h(x) = 1.8x + 32$ is the temperature in °F.

$E(h(x)) = 1.8\, E(x) + 32 = 248.$

$V(h(x)) = 1.8^2 \cdot V(x) = 12.96.$

Thus $\sigma(h(x)) = \sqrt{V(h(x))} = 3.6.$